EXPERIMENT NO. 9 SOFTWARE LICENSE COMPLIANCE AUDIT AND NETWORK VULNERABILITY SCANNING

AIM:

To audit software and system information using WinAudit and identify open ports, running services, and potential vulnerabilities using Nmap Vulnerability Engine.

OBJECTIVES:

1. To examine system configuration and software-related information using WinAudit and understand its role in software inventory auditing.
2. To understand the importance of software license compliance and the need to verify installed software against applicable license terms.
3. To install Nmap on a Windows system and verify its successful installation using the Command Prompt.
4. To identify open ports and running services on the local computer and understand their significance in network security.
5. To execute Nmap vulnerability scripts, analyze the scan results, and understand basic preventive measures to improve system security.

HARDWARE & SOFTWARE REQUIREMENTS:

Category	Specification & Tools
Hardware Configuration	PC/Laptop with Intel Core i5 or higher, minimum 4 GB RAM, Internet Connection
Operating System	Windows 10/11
Software Audit Tool	WinAudit
Network Scanning Tool	Nmap 7.99
Packet Capture Utility	Npcap 1.88
Command-Line Utility	Windows Command Prompt
Documentation Tools	Microsoft Word

THEORY:

Software license compliance auditing is the process of examining installed software and available licensing information to determine whether software usage follows the applicable license terms. It helps organizations identify unauthorized software, duplicate installations, expired licenses, and possible licensing violations. WinAudit is a system auditing utility that collects information about the computer's hardware, operating system, installed software, and other system settings. This information supports software inventory auditing and helps administrators understand the software present on a system. However, complete license compliance verification also requires checking license agreements, activation records, and proof of purchase where applicable.

Nmap (Network Mapper) is an open-source network scanning tool used to discover active hosts, identify open ports, and detect services running on a computer or network. It helps administrators understand which services are accessible and identify ports that may require further security review. Nmap supports different scanning options, including port scanning and service version detection, which provide useful information for assessing network security.

Nmap includes a Network Scripting Engine (NSE) that uses scripts to perform additional network checks. The command nmap --script vuln runs scripts from the vulnerability category against a specified target. These scripts check for selected known vulnerabilities and configuration weaknesses. The command can help identify potential security issues that may require further investigation.

The scan results must be interpreted carefully. An open port indicates that a service is listening on that port, but it does not automatically mean the service is vulnerable. Similarly, a script result indicating that a vulnerability was not detected does not guarantee that the system is completely secure. Some checks may fail because of connection errors, unsupported services, or scanning limitations. Therefore, scan findings should be reviewed and verified before drawing conclusions.

By combining WinAudit and Nmap, system auditing and network vulnerability assessment can be performed as complementary security activities. WinAudit helps collect system and software information, while Nmap helps examine network ports, services, and selected security weaknesses. These activities support better system management, early identification of potential risks, and the implementation of appropriate preventive measures.

ANALYSIS OF SOFTWARE LICENSE COMPLIANCE AUDIT AND NETWORK VULNERABILITY SCANNING – PHASES, TOOLS AND COMMANDS:

1. **Software License Compliance Audit Using WinAudit**
   In this phase, WinAudit is used to collect system configuration and software-related information. The System Overview and Active Setup sections are examined to understand the computer's configuration and available software details.

Tool / Utility	Purpose	Action Used
WinAudit	Collect system information	Open WinAudit
System Overview	Examine hardware and operating system details	Select System Overview
Installed Software	Access software-related information	Navigate to Installed Software
Active Setup	Examine Active Setup details	Select Active Setup

2. **Network Vulnerability Scanning Using Nmap**
   In this phase, Nmap is used to identify open ports, detect running services, and check for selected security vulnerabilities on the local computer. The scan results are examined to understand potential network security risks.

Tool / Utility	Purpose	Command Used
Command Prompt	Verify Nmap installation	nmap --version
Nmap NSE	Check for selected vulnerabilities	nmap --script vuln 127.0.0.1
Nmap Service Detection	Detect service versions and scan selected ports	nmap -sV --script vuln -p 1-1024 127.0.0.1

IMPLEMENTATION STEPS:

1. **Software License Compliance Audit Using WinAudit**
   - Opened WinAudit on the Windows system to collect system configuration and software-related information.
   - Examined the System Overview section to identify hardware and operating system details.
   - Navigated to the Installed Software section to access software-related information.

- Reviewed the Active Setup section to understand the available software configuration details.
- Examined the collected information to understand its role in software inventory auditing and license compliance verification.

2. **Network Vulnerability Scanning Using Nmap**
   - Verified the Nmap installation by executing the nmap --version command.
   - Executed nmap --script vuln 127.0.0.1 to check the local computer for selected known vulnerabilities.
   - Executed nmap -sV --script vuln -p 1-1024 127.0.0.1 to detect service versions and perform vulnerability checks on the specified TCP ports.
   - Examined the scan results to identify open ports, running services, and reported vulnerability findings.
   - Reviewed the output to understand potential security risks, noting that some vulnerability checks returned connection errors and were inconclusive.

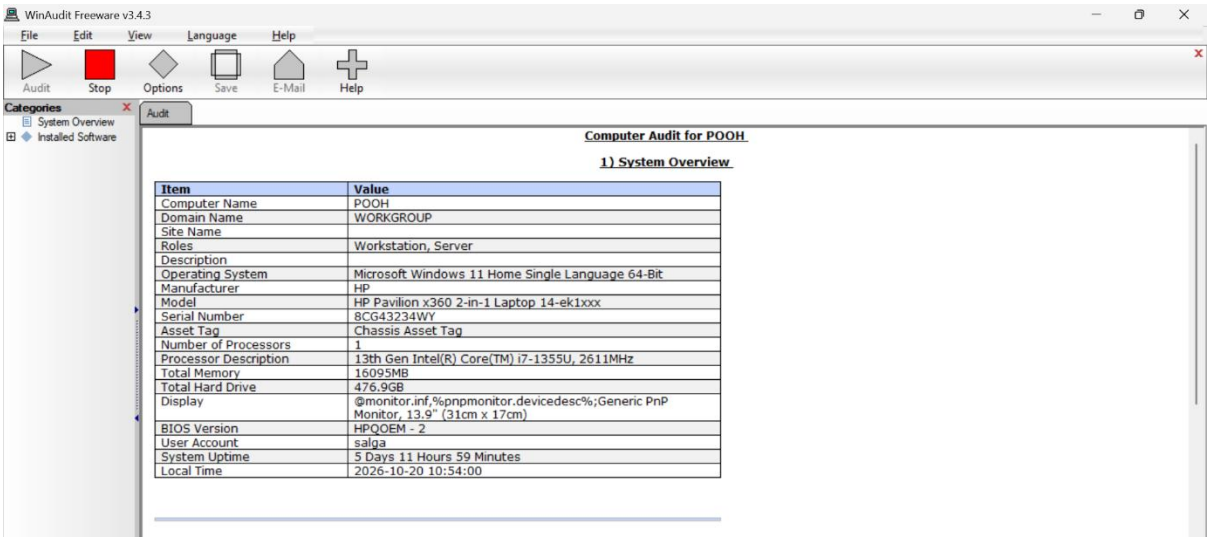
OBSERVATIONS:

During the cybersecurity practical, WinAudit and Nmap were used to examine system configuration, software-related information, open ports, and running services. WinAudit was used to review hardware, operating system, and software-related details, while Nmap was used to verify installation and scan the local computer for potential vulnerabilities. The results helped understand possible security risks and the importance of system auditing and network security.

- WinAudit displayed the system configuration, including hardware and operating system details.
- The Active Setup section under Installed Software was examined to understand the available software-related information.
- The collected information demonstrated how system auditing can support software inventory and license compliance verification.
- Nmap installation was verified successfully using the nmap --version command.
- The command nmap --script vuln 127.0.0.1 was executed to check for selected known vulnerabilities on the local computer.
- Open ports and running services were identified during the Nmap scan.
- Some SMB vulnerability checks returned connection negotiation errors, while the displayed results did not confirm a vulnerability.

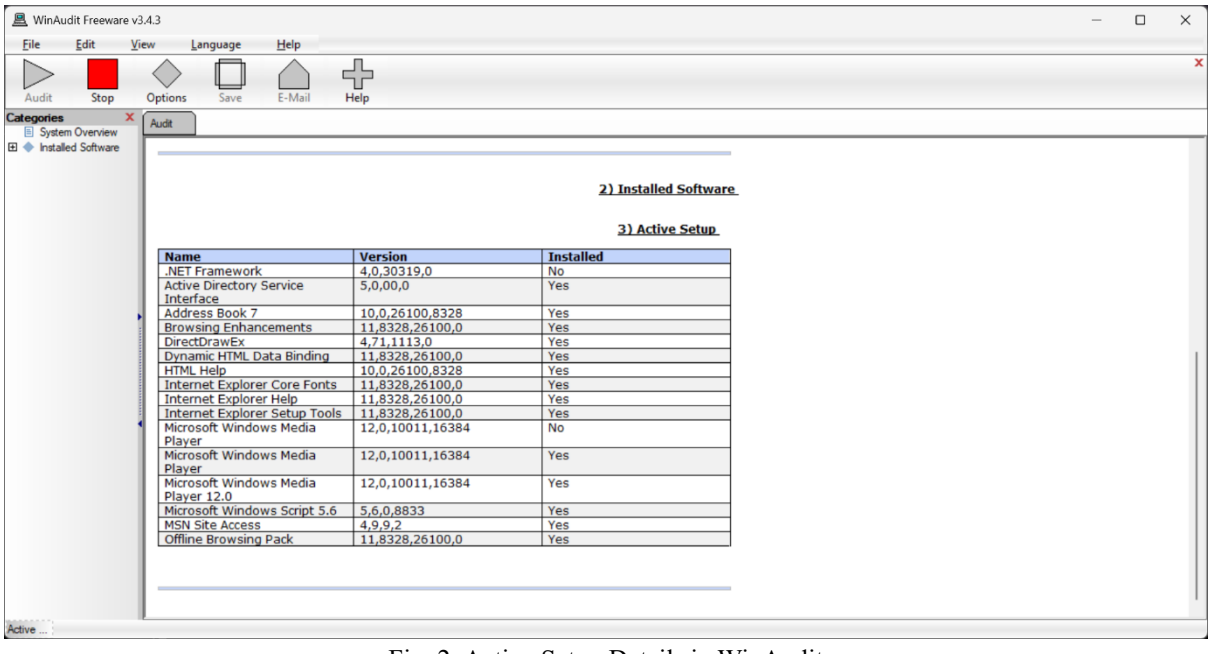
OUTPUT:

Phase 1 – Software License Compliance Audit Using WinAudit



Item	Value
Computer Name	POOH
Domain Name	WORKGROUP
Site Name	
Roles	Workstation, Server
Description	
Operating System	Microsoft Windows 11 Home Single Language 64-Bit
Manufacturer	HP
Model	HP Pavilion x360 2-in-1 Laptop 14-ek1xxx
Serial Number	8CG43234WY
Asset Tag	Chassis Asset Tag
Number of Processors	1
Processor Description	13th Gen Intel(R) Core(TM) i7-1355U, 2611MHz
Total Memory	16095MB
Total Hard Drive	476.9GB
Display	@monitor.inf,%pnpmonitor.devicedesc%;Generic PnP Monitor, 13.9" (31cm x 17cm)
BIOS Version	HPQOEM - 2
User Account	salga
System Uptime	5 Days 11 Hours 59 Minutes
Local Time	2026-10-20 10:54:00

Fig. 1: System Configuration Audit Using WinAudit



Name	Version	Installed
.NET Framework	4,0,30319,0	No
Active Directory Service Interface	5,0,00,0	Yes
Address Book 7	10,0,26100,8328	Yes
Browsing Enhancements	11,8328,26100,0	Yes
DirectDrawEx	4,71,1113,0	Yes
Dynamic HTML Data Binding	11,8328,26100,0	Yes
HTML Help	10,0,26100,8328	Yes
Internet Explorer Core Fonts	11,8328,26100,0	Yes
Internet Explorer Help	11,8328,26100,0	Yes
Internet Explorer Setup Tools	11,8328,26100,0	Yes
Microsoft Windows Media Player	12,0,10011,16384	No
Microsoft Windows Media Player	12,0,10011,16384	Yes
Microsoft Windows Media Player 12.0	12,0,10011,16384	Yes
Microsoft Windows Script 5.6	5,6,0,8833	Yes
MSN Site Access	4,9,9,2	Yes
Offline Browsing Pack	11,8328,26100,0	Yes

Fig. 2: Active Setup Details in WinAudit

Phase 2 – Network Vulnerability Scanning Using Nmap

```
C:\Users\salga>nmap --version
Nmap version 7.99 ( https://nmap.org )
Platform: i686-pc-windows-windows
Compiled with: nmap-liblua-5.4.8 openssl-3.0.16 nmap-libssh2-1.11.1 nmap-libz-1.3.2 nmap-libpcre2-10.47 Npcap-1.88 nmap-libdnet-1.18.
0 ipv6
Compiled without:
Available nsock engines: iocp poll select
```

Fig. 3: Nmap Version Verification

```
C:\Users\salga>nmap --script vuln 127.0.0.1
Starting Nmap 7.99 ( https://nmap.org ) at 2026-10-20 11:02 +0530
Nmap scan report for localhost (127.0.0.1)
Host is up (0.000098s latency).
Not shown: 996 closed tcp ports (reset)
PORT     STATE SERVICE
21/tcp   open  ftp
135/tcp  open  msrpc
445/tcp  open  microsoft-ds
5432/tcp open  postgresql

Host script results:
|_samba-vuln-cve-2012-1182: Could not negotiate a connection:SMB: Failed to receive bytes: ERROR
|_smb-vuln-ms10-054: false
|_smb-vuln-ms10-061: Could not negotiate a connection:SMB: Failed to receive bytes: ERROR

Nmap done: 1 IP address (1 host up) scanned in 71.75 seconds
```

Fig. 4: Nmap Vulnerability Scan on Localhost

```
C:\Users\salga>nmap -sV --script vuln -p 1-1024 127.0.0.1
Starting Nmap 7.99 ( https://nmap.org ) at 2026-10-20 11:07 +0530
Nmap scan report for localhost (127.0.0.1)
Host is up (0.0012s latency).
Not shown: 1020 closed tcp ports (reset)
PORT    STATE    SERVICE       VERSION
21/tcp  open     ftp           Microsoft ftpd
135/tcp open     msrpc         Microsoft Windows RPC
137/tcp filtered netbios-ns
445/tcp open     microsoft-ds?
Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows

Host script results:
|_smb-vuln-ms10-054: false
|_smb-vuln-ms10-061: Could not negotiate a connection:SMB: Failed to receive bytes: ERROR
|_samba-vuln-cve-2012-1182: Could not negotiate a connection:SMB: Failed to receive bytes: ERROR

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 47.57 seconds
```

Fig. 5: Service Detection and Vulnerability Scan Using Nmap

ANALYSIS:

The experiment successfully demonstrated software auditing using WinAudit and network vulnerability scanning using Nmap. System configuration and software-related information were examined, while Nmap was used to identify open ports, running services, and potential vulnerabilities. Some SMB checks returned connection errors, and no vulnerability was confirmed in the displayed results. The practical highlighted the importance of reviewing scan results carefully to identify potential security risks.

PREVENTIVE MEASURES:

1. Keep the Windows operating system and installed applications updated with the latest security patches.
2. Verify installed software against applicable license terms and maintain valid license records.
3. Disable unnecessary services and close ports that are not required.
4. Use firewall rules to restrict unauthorized access to network services.
5. Run regular vulnerability scans to identify potential security weaknesses.
6. Investigate inconclusive scan results and verify suspected vulnerabilities.
7. Review system configuration and software inventory regularly to maintain system security.

CONCLUSION:

The experiment successfully demonstrated software auditing using WinAudit and network vulnerability scanning using Nmap. System configuration, software-related information, open ports, and running services were examined to understand their security significance. The practical helped develop an understanding of software inventory auditing and vulnerability assessment. It also highlighted the importance of regular audits, security updates, and preventive measures to reduce potential security risks.